

Naive Bayes Classifier example

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1 The Classifier

The Bayes Naive classifier selects the most likely classification V_{nb} given the attribute values $a_1, a_2, \dots, a_n$. This results in:

$$V_{nb} = \operatorname{argmax}_{v_j \in V} P(v_j) \prod P(a_i|v_j) \quad (1)$$

We generally estimate $P(a_i|v_j)$ using m-estimates:

$$P(a_i|v_j) = \frac{n_c + mp}{n + m} \quad (2)$$

where:

- n = the number of training examples for which $v = v_j$
- n_c = number of examples for which $v = v_j$ and $a = a_i$
- p = a priori estimate for $P(a_i|v_j)$
- m = the equivalent sample size

2 Car theft Example

Attributes are Color , Type , Origin, and the subject, stolen can be either yes or no.

2.1 data set

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

2.2 Training example

We want to classify a Red Domestic SUV. Note there is no example of a Red Domestic SUV in our data set. Looking back at equation (2) we can see how to compute this. We need to calculate the probabilities

$P(\text{Red}|\text{Yes})$, $P(\text{SUV}|\text{Yes})$, $P(\text{Domestic}|\text{Yes})$,

$P(\text{Red}|\text{No})$, $P(\text{SUV}|\text{No})$, and $P(\text{Domestic}|\text{No})$

and multiply them by P(Yes) and P(No) respectively . We can estimate these values using equation (3).

Yes :

Red :

n = 5
n_c = 3
p = .5
m = 3

SUV :

n = 5
n_c = 1
p = .5
m = 3

Domestic :

n = 5
n_c = 2
p = .5
m = 3

No :

Red :

n = 5
n_c = 2
p = .5
m = 3

SUV :

n = 5
n_c = 3
p = .5
m = 3

Domestic :

n = 5
n_c = 3
p = .5
m = 3

Looking at $P(Red|Yes)$, we have 5 cases where $v_j = Yes$, and in 3 of those cases $a_i = Red$. So for $P(Red|Yes)$, $n = 5$ and $n_c = 3$. Note that all attribute are binary (two possible values). We are assuming no other information so, $p = 1 / (\text{number-of-attribute-values}) = 0.5$ for all of our attributes. Our m value is arbitrary, (We will use $m = 3$) but consistent for all attributes. Now we simply apply equation (3) using the precomputed values of n , n_c , p , and m .

$$\begin{aligned} P(Red|Yes) &= \frac{3 + 3 * .5}{5 + 3} = .56 & P(Red|No) &= \frac{2 + 3 * .5}{5 + 3} = .43 \\ P(SUV|Yes) &= \frac{1 + 3 * .5}{5 + 3} = .31 & P(SUV|No) &= \frac{3 + 3 * .5}{5 + 3} = .56 \\ P(Domestic|Yes) &= \frac{2 + 3 * .5}{5 + 3} = .43 & P(Domestic|No) &= \frac{3 + 3 * .5}{5 + 3} = .56 \end{aligned}$$

We have $P(Yes) = .5$ and $P(No) = .5$, so we can apply equation (2). For $v = Yes$, we have

$$P(Yes) * P(Red | Yes) * P(SUV | Yes) * P(Domestic|Yes)$$

$$= .5 * .56 * .31 * .43 = .037$$

and for $v = No$, we have

$$P(No) * P(Red | No) * P(SUV | No) * P(Domestic | No)$$

$$= .5 * .43 * .56 * .56 = .069$$

Since $0.069 > 0.037$, our example gets classified as 'NO'